Full Survey Instrument (English Version)

Items and Scale Development

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Supplementary Materials

Note

The survey instrument was originally developed in English. For the experimental implementation, the survey was translated into Spanish, which served as the primary language of data collection. The English version presented in this document reflects the original item formulations and serves as the reference version for documentation and scale development. All measurement items are identical in meaning across both language versions. Minor differences in instructional or contextual phrasing may exist between versions; these do not affect the measurement items and were harmonized to ensure full conceptual equivalence.

In the actual survey implementation, items were presented in separate blocks, consistent with the original Spanish instrument (see Appendix: *06_Survey_Interface_Qualtrics_ES*). This design choice was made to optimize usability on mobile devices, as a substantial proportion of participants were expected to complete the survey via smartphone. In particular, presenting Likert-scale items in smaller blocks improves readability and reduces cognitive load in mobile survey interfaces.

For the same reason, all scale points were explicitly labeled, as endpoint-only labeling can reduce clarity and increase response error in mobile-based survey formats.

The present document reproduces the full survey in its original English version. All items are labeled according to their corresponding Qualtrics variable names. Section headers have been slightly adapted to improve clarity from the perspective of the experimental structure, without altering the underlying constructs or item meanings.

Details on the development and adaptation of the employed scales are provided in Chapter 5, including methodological justifications for item selection, modification, and scale construction.

For the original survey instrument, please refer to the Appendix *"06_Survey_Interface_Qualtrics_ES"*.

1. Pre-Intervention Survey

Survey: Interaction with an NGO Chatbot

Hello,

Thank you for participating in this study!

My name is Joël Sieber and I am a Master's student at the University of St. Gallen (HSG), Switzerland. This study aims to understand how people interact with digital support tools of non-governmental organizations (NGOs).

- Your participation is completely **voluntary**.
- You can **leave the study at any time**.
- Your responses are **anonymous** and confidential.
- The scenarios are **hypothetical**: you do not need to provide real data.

Completing the survey will take approximately **11 minutes**. If you have any questions: joel.sieber@student.unisg.ch

By clicking "**Continue**," you confirm that you are at least **18 years old** and agree to participate voluntarily.

Section 1: Inclusion / Screening Questions

DEM_age: **What's your age?**

___ years.

DEM_medellin: **Are you enrolled at a public university in Medellín?**

◯ Yes
◯ No

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2. Redirect to Intervention

Section 2: Scenario & Chatbot Interaction

Instructions

You will now interact with a chatbot used by an NGO in a hypothetical support scenario. Please read the introduction carefully and proceed with "Next". Then, read the messages carefully and respond naturally, as if this were a real interaction. Please do **not refresh the page** during the interaction.

At the end of the conversation, click on "Continue" to complete the survey.

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3. Post-Intervention Survey

Section 3: Scenario Immersion & Manipulation Check

Now we will ask a few questions about how the interaction came across.

3.1 Scenario Immersion - adapted from Seitz (2024, p. 12)

IMM: Please indicate how much you agree with the following statements.

	1. Strongly disagree	2. Disagree	3. Somewhat disagree	4. Neither agree nor disagree	5. Somewhat agree	6. Agree	7. Strongly agree
IMM_01: **The chatbot could exist in reality.**	○	○	○	○	○	○	○
IMM_02: **I was able to imagine the situation very well.**	○	○	○	○	○	○	○
IMM_03: **The interaction felt realistic.**	○	○	○	○	○	○	○
IMM_04: **Overall, the scenario was credible.**	○	○	○	○	○	○	○

3.2 MC: Empathy & Recognition - Self-developed

EMP: Please indicate how much you agree with the following statements.

	1. Strongly disagree	2. Disagree	3. Somewhat disagree	4. Neither agree nor disagree	5. Somewhat agree	6. Agree	7. Strongly agree
EMP_01: **The chatbot acknowledged emotional aspects of the situation.**	○	○	○	○	○	○	○
EMP_02: **The chatbot responded in a warm and understanding way.**	○	○	○	○	○	○	○
EMP_03: **The chatbot appeared attentive to how the situation might feel.**	○	○	○	○	○	○	○

REC: **Please indicate how much you agree with the following statements.**

	1. Strongly disagree	2. Disagree	3. Somewhat disagree	4. Neither agree nor disagree	5. Somewhat agree	6. Agree	7. Strongly agree
REC_01: **The chatbot emphasized that what happens next depends on me.**	○	○	○	○	○	○	○
REC_02: **The chatbot structured the interaction in a way that kept the decision in my hands.**	○	○	○	○	○	○	○
REC_03: **The chatbot communicated clear boundaries about its role (what it can vs. cannot do).**	○	○	○	○	○	○	○

Section 4: Disclosure Willingness

4. Self-reported disclosure willingness (secondary validation) - Self-developed

DISC: **Please indicate how much you agree with the following statements.**

	1. Strongly disagree	2. Disagree	3. Somewhat disagree	4. Neither agree nor disagree	5. Somewhat agree	6. Agree	7. Strongly agree
DISC_01: **I would have been willing to share personal information with this chatbot if needed.**	○	○	○	○	○	○	○
DISC_02: **Providing personal details in this interaction felt acceptable to me.**	○	○	○	○	○	○	○
DISC_03: **I would feel comfortable sharing sensitive information with this chatbot.**	○	○	○	○	○	○	○

Section 5: Primary Outcomes

The following statements describe how the chatbot interacted with you.

5.1 Perceived Paternalism - adapted from Williams et al. (2012)

PAT: Please indicate how much you agree with the following statements.

	1. Strongly disagree	2. Disagree	3. Somewhat disagree	4. Neither agree nor disagree	5. Somewhat agree	6. Agree	7. Strongly agree
PAT_01: **The chatbot treated me as less capable than I am.**	○	○	○	○	○	○	○
PAT_02: **The chatbot behaved in a controlling manner.**	○	○	○	○	○	○	○
PAT_03: **The chatbot tried to steer me toward a particular choice rather than helping me decide.**	○	○	○	○	○	○	○
PAT_04: **The chatbot's guidance felt overly directive for its role.**	○	○	○	○	○	○	○
PAT_05: **The chatbot seemed to overstep what is appropriate for an NGO chatbot.**	○	○	○	○	○	○	○

5.2 Affective & Cognitive Trust - adapted from Shang et al. (2024, p. 1348)

CTRU/ATRU: **Please rate the chatbot based on the following pairs of descriptors.**

		-3	-2	-1	0	+1	+2	+3	
CTRU_01	**Clueless**	○	○	○	○	○	○	○	**Knowledgeable**
CTRU_02	**Incompetent**	○	○	○	○	○	○	○	**Competent**
CTRU_03	**Ineffective**	○	○	○	○	○	○	○	**Effective**
CTRU_04	**Inexperienced**	○	○	○	○	○	○	○	**Experienced**
ATRU_01	**Rude**	○	○	○	○	○	○	○	**Cordial**
ATRU_02	**Indifferent**	○	○	○	○	○	○	○	**Responsive**
ATRU_03	**Judgmental**	○	○	○	○	○	○	○	**Open-minded**
ATRU_04	**Impatient**	○	○	○	○	○	○	○	**Patient**

5.3 Perceived User Agency - self developed

AGE: **Please indicate how much you agree with the following statements.**

	1. Strongly disagree	2. Disagree	3. Somewhat disagree	4. Neither agree nor disagree	5. Somewhat agree	6. Agree	7. Strongly agree
AGE_01: **I experienced myself as the one making the decisions during the interaction.**	○	○	○	○	○	○	○
AGE_02: **I felt free to decide how to proceed.**	○	○	○	○	○	○	○
AGE_03: **I did not feel pushed toward a particular option.**	○	○	○	○	○	○	○
AGE_04: **I experienced the interaction as respecting my ability to decide for myself.**	○	○	○	○	○	○	○

Section 6: Control Constructs & Attention Check (ATTCH)

6.1 Social Presence - adapted from Munnukka et al. (2022, p. 6)

SOC: **Please indicate how much you agree with the following statements.**

	1. Strongly disagree	2. Disagree	3. Somewhat disagree	4. Neither agree nor disagree	5. Somewhat agree	6. Agree	7. Strongly agree
SOC_01: **I felt a sense of human contact with the chatbot.**	○	○	○	○	○	○	○
SOC_02: **I felt a sense of personalness with the chatbot.**	○	○	○	○	○	○	○
ATTCH: **Please select "Somewhat agree".**	○	○	○	○	○	○	○
SOC_04: **I felt a sense of human sensitivity with the chatbot.**	○	○	○	○	○	○	○

6.2 Perceived Privacy Risk - Cheng et al. (2025)

RISK: **Please indicate how much you agree with the following statements.**

	1. Strongly disagree	2. Disagree	3. Somewhat disagree	4. Neither agree nor disagree	5. Somewhat agree	6. Agree	7. Strongly agree
RISK_01: **My decision to use the chatbot involves a certain level of risk.**	○	○	○	○	○	○	○
RISK_02 (reverse scored): **The security systems built into the chatbot may be strong enough to protect my personal information.**	○	○	○	○	○	○	○
RISK_03: **I do not feel totally safe providing personal private information over chatbots.**	○	○	○	○	○	○	○

Section 7: Background (Exploratory or Covariates)

Finally, we ask a few general background questions.

7.1 Prior Chatbot Use - adapted from Cheng et al. (2025, p. 3919)

EXP_chatbot: **How often have you used any types of chatbots (e.g. ChatGPT, WhatsApp Chatbots, Google's Gemini) before?**

◯ 1. Never
◯ 2. Once or twice
◯ 3. A few times per year
◯ 4. About once per month
◯ 5. About once per week
◯ 6. Several times per week
◯ 7. Daily or almost daily

7.2 Prior NGO Experience - self developed

EXP_ngo_binary: **Have you had personal contact with an NGO before? – self developed**

◯ No
◯ Yes

EXP_ngo_type: **(If yes) What describes your most common contact with NGOs? – self developped**

◯ 1. Received support
◯ 2. Volunteered
◯ 3. Worked / Interned (paid or unpaid)
◯ 4. Donor / member
◯ 5. Other

7.3 General Trust in NGO - self developed

TRU_ngo: **Please indicate how much you agree with the following statements.**

	1. Strongly disagree	2. Disagree	3. Somewhat disagree	4. Neither agree nor disagree	5. Somewhat agree	6. Agree	7. Strongly agree
TRU_ngo_01: **I trust NGOs generally.**	◯	◯	◯	◯	◯	◯	◯
TRU_ngo_02: **NGOs usually act in the public interest.**	◯	◯	◯	◯	◯	◯	◯
TRU_ngo_03: **NGOs can generally be relied upon to do what is right.**	◯	◯	◯	◯	◯	◯	◯

Section 8: Demographics

DEM_gender: **Gender:**

◯ Male
◯ Female
◯ Diverse
◯ Prefer not to say

P1: **In your opinion, what do you think was the main purpose of this study? (If you have any idea, briefly describe it.)**

_____

P2: **Do you have any comment or suggestion about the study? (This question is optional.)**

_____

MAIL: **Please enter your student email address here if you wish to participate in the raffle. This will be stored separately; your responses will have no impact on the lottery. Then, on the next page, click "Submit" so that your responses and your email address are recorded.**

_____

End Screen

Final Study Information

Your contribution is essential for this research.

Objective: The study examines how the language style of a chatbot influences trust in NGOs.

Nature of the study: This study involved an academic simulation. The chatbot did not represent a real organization, and all data presented were hypothetical.

Privacy: No personally identifiable data were collected. Your responses are 100% anonymous.

If you have questions or wish to withdraw your data, contact:
Joël Sieber (HSG)
Email: joel.sieber@student.unisg.ch

Click the button below to finish and submit your responses.

4. Original Scales

3.1 Scenario Immersion – adapted from Seitz (2024, p. 12)

To measure scenario immersion, four items were adapted from Seitz (2024), originally developed by Gelbrich et al. (2021). The original wording (focusing on an interaction "between the chatbot and patient") was modified to reflect the participant's own interaction in the scenario.

- **Original Items (Seitz, 2024):** (1) The chatbot could exist in reality, (2) I was able to imagine the situation very well, (3) The interaction *between the chatbot and patient was* realistic, (4) Overall, the scenario was credible.
- **Final Items (used in this study):** (1) The chatbot could exist in reality, (2) I was able to imagine the situation very well, (3) The interaction felt realistic, (4) Overall, the scenario was credible**.**
- **Measurement:** 7-point Likert scale (1 = Strongly disagree to 7 = Strongly agree).
- **Reliability:** Seitz' (2024) Cronbach's Alpha (α): 0.93

5.1 Perceived Paternalism – adapted from Williams et al. (2012)

Perceived paternalism was measured using five items adapted from the "Control/Dominance" subscale of the Emotional Tone Rating Scale (ETRS) developed by Williams et al. (2012, p. 382). While the full ETRS consists of 12 items (including a "Care/Affection" dimension), only the five items loading on the "Control/Dominance" factor were selected, as they directly operationalize the restrictive and directive nature of paternalistic communication.

- **Original Items (Control/Dominance Subscale):** The original scale uses five adjective-based semantic descriptors: *Directive, Patronizing, Bossy, Dominating,* and *Controlling*.
- **Adaptation:** The original 5-point scale (1 = "Not at all" to 5 = "Very") was converted to a 7-point Likert scale (1 = "Strongly disagree" to 7 = "Strongly agree") to ensure consistency with other measures in this study. The adjectives were rephrased into full-sentence statements to assess the chatbot's behavior.
- **Final Items (used in this study):** (1) The chatbot treated me as less capable than I am, (2) The chatbot behaved in a controlling manner, (3) The chatbot tried to steer me toward a particular choice rather than helping me decide, (4) The chatbot's guidance felt overly directive for its role, (5) The chatbot seemed to overstep what is appropriate for an NGO chatbot.
- **Measurement:** 7-point Likert scale (1 = Strongly disagree to 7 = Strongly agree).
- **Reliability:** The "Control/Dominance" subscale demonstrated high internal consistency in the original psychometric analysis with a Cronbach's alpha of $\alpha = .94$ (Williams et al., 2012, p. 383).

5.2 Affective & Cognitive Trust – Shang et al. (2024, p. 1348)

Affective and cognitive trust was measured using a parsimonious 8-item adaptation of the Semantic Differential Scale for AI Trust (Shang et al., 2024, p. 1348) to measure trust in the NGO-chatbot interaction while minimizing respondent fatigue. The selection focuses on two core sub-dimensions based on the highest factor loadings (λ) and internal consistencies (α) reported in the original validation:

Cognitive Trust (CTRU) – Sub-dimension: Competence

- Items: Knowledgeable (λ = .91), Competent (λ = .90), Effective (λ = .86), and Experienced (λ = .75).
- Original Reliability: α = .98. This dimension reflects the agent's perceived ability to provide accurate support.

Affective Trust (ATRU) – Sub-dimension: Amiability

- Items: Cordial (λ = .99), Responsive (λ = .71), Open-minded (λ = .69), and Patient (λ = .58).
- Original Reliability: α = .96. This dimension captures the relational tone of the interaction.

The original Benevolence sub-scale (e.g., Caring, Altruistic) was intentionally excluded to ensure discriminant validity against the Perceived Paternalism construct (Williams et al., 2012). Since paternalism is conceptually linked to "misguided benevolence," focusing the trust measure on Amiability (tone) rather than intent prevents multicollinearity and ensures a clean statistical separation between perceived friendliness and controlling behavior. Following Shang et al. (2024), a 7-point bipolar semantic differential (-3 to +3) was used. This format preserves the original psychometric properties and provides a distinct evaluative frame, differentiating it from the 7-point Likert scales used for other constructs.

Figure 1
Initial Scale by Shang et al., 2024, p. 1348

			All		AI Condition		
#	Item	Sub-dimension	F1	F2	F1	F2	Source
C1	Unreliable - Reliable	Reliability	0.905	0	0.922	-0.058	a,b,c,d,e
C2	Inconsistent - Consistent		0.928	-0.12	0.924	-0.183	a,b,c,d
C3	Unpredictable - Predictable		0.894	-0.171	0.898	-0.204	d
C4	Undependable - Dependable		0.851	0.016	0.911	-0.111	a,b,c,d
C5	Fickle - Dedicated		0.759	0.139	0.744	0.116	a,b,d
C6	Careless - Careful		0.721	0.213	0.716	0.206	a,b
C7	Unbelievable - Believable		0.69	0.082	0.658	0.034	c,d,e
C8	Unpromising - Promising						c,d,e
C9	Clueless - Knowledgable	Competence	0.907	-0.018	0.898	0.026	f,d,e
C10	Incompetent - Competent		0.9	0.023	0.921	-0.021	a,b,f,d,e
C11	Ineffective - Effective		0.861	0.075	0.863	0.071	d,e
C12	Inexperienced - Experienced		0.751	0.089	0.611	0.155	a,b,f,e
C13	Amateur - Proficient	Understandability	0.895	0.009	0.864	0.039	a,b,c,f,d,e
C14	Irrational - Rational		0.827	0.02	0.792	0.05	d,e
C15	Unreasonable - Reasonable		0.71	0.224	0.714	0.202	d,e
C16	Incomprehensible - Understandable		0.706	0.175	0.783	0.079	f,d
C17	Opaque - Transparent		0.6	0.261	0.656	0.18	c,f,d
C18	Dishonest - Honest	Integrity	0.693	0.178	0.743	0.097	c,f,e
C19	Unfair - Fair		0.663	0.274	0.66	0.268	c,f
C20	Insincere - Sincere						f,e
A1	Apathetic - Empathetic	Benevolence	-0.11	0.989	-0.162	0.967	a,b
A2	Insensitive - Sensitive		-0.08	0.959	-0.109	0.955	a,b,c
A3	Impersonal - Personal		-0.024	0.902	-0.025	0.847	b,c,d
A4	Ignoring - Caring		0.048	0.881	-0.055	0.941	a,b,c
A5	Self-serving - Altruistic		0.215	0.627	0.207	0.622	b,c,e
A6	Malicious - Benevolent						f,e
A7	Harmful - Well-intentioned						f,e
A8	Discouraging - Supportive						c,f,e
A9	Rude - Cordial	Amiability	-0.11	0.989	0.112	0.76	a,b
A10	Indifferent - Responsive		0.221	0.711	0.232	0.667	a,b,c,f,e
A11	Judgemental - Open-minded		0.142	0.688	0.078	0.697	c
A12	Impatient - Patient		0.291	0.577	0.218	0.6	a,b
A13	Unpleasant - Likable						f,d

5.3 Perceived User Agency – self developed

Perceived User Agency was measured using a four-item scale specifically constructed for this study to capture participants' subjective experience of decision authority during the interaction. The items focus on whether participants experienced themselves as autonomous decision-makers and whether the interaction respected their ability to choose how to proceed.

The scale was designed to measure interaction-level agency rather than general self-efficacy, empowerment, or motivational states. Items therefore emphasize perceived freedom of choice and the absence of pressure, without referencing emotional support, competence, or personal development.

- **Final Items (used in this study):** (1) I experienced myself as the one making the decisions during the interaction, (2) I felt free to decide how to proceed, (3) I did not feel pushed toward a particular option (4) I experienced the interaction as respecting my ability to decide for myself.
- **Measurement:** 7-point Likert scale (1 = Strongly disagree to 7 = Strongly agree).

Perceived User Agency is **treated as a secondary, descriptive outcome and** is not modeled as an independent mechanism or mediator in the confirmatory analyses.

6.1 Social Presence – adapted from Munnukka et al. (2022, p. 6)

Social presence was measured using three items adapted from Munnukka et al. (2022, p. 6), who originally utilized a five-item scale developed by Verhagen et al. (2014, p. 538). This construct captures the "sense of being with another" during the digital interaction.

- **Original Items (Munnukka et al., 2022**): The original scale used by Munnukka et al. includes the following three items: (1) I felt a sense of human contact with the VSA, (2) I felt a sense of personalness with the VSA, (3) I felt a sense of human sensitivity with the VSA.
- **Adaptation:** The term "VSA" (Virtual Service Assistant) from the original study was replaced with "chatbot" to align with the terminology used in this experiment's scenario.
- **Final Items (used in this study):** (1) I felt a sense of human contact with the chatbot, (2) I felt a sense of personalness with the chatbot, (3) I felt a sense of human sensitivity with the chatbot.
- **Measurement:** 7-point Likert scale (1 = Strongly disagree to 7 = Strongly agree).
- **Reliability:** In the psychometric analysis by Munnukka et al. (2022), the social presence scale demonstrated high internal consistency with a composite reliability (CR) of 0.884 and an average variance extracted (AVE) of 0.719.

6.2 Perceived Privacy Risk – adapted from Cheng et al. (2025)

Perceived privacy risk was measured using three items adapted from Cheng et al. (2025, p. 3919), who originally adapted the items from Kasilingam (2020, p. 12). This scale assesses participants' concerns regarding data security and the potential misuse of personal information when interacting with a chatbot.

- **Original Items (Cheng et al., 2025):** The original scale utilized by Cheng et al. includes the following three items: (1) My decision to use the chatbot involves a certain level of risk, (2) The security systems built into the chatbot may be strong enough to protect my personal information (reverse scored), (3) I do not feel totally safe providing personal private information over chatbots.

- **Adaptation:** The items were used in their established form to measure the inhibiting effect of privacy risks on user satisfaction. The measurement was adjusted to a 7-point scale for consistency across this study's instruments.

- **Final Items (used in this study**): (1) My decision to use the chatbot involves a certain level of risk, (2) The security systems built into the chatbot may be strong enough to protect my personal information (3) I do not feel totally safe providing personal private information over chatbots.

- **Measurement:** 7-point Likert scale (1 = Strongly disagree to 7 = Strongly agree).

- **Reliability:** In the study by Cheng et al. (2025), the privacy risk scale demonstrated good internal consistency with a Cronbach's alpha of 0.79.

7.1 Prior Chatbot Use – adapted from Cheng et al. (2025, p. 3919)

Prior chatbot experience was measured using a single-item frequency scale adapted from Cheng et al. (2025, p. 3919). This variable serves as a covariate to control for the participants' general familiarity with conversational AI.

- **Original Item (Cheng et al., 2025):** " How often have you used any types of chatbots (e.g., ChatGPT, Apple's Siri, Amazon's Alexa, Facebook Messenger chatbots, WhatsApp chatbots, and Google Assistant) before participating in this survey?"

- **Adaptation:** The examples provided in the prompt were updated to reflect contemporary LLM-based tools (*ChatGPT, WhatsApp Chatbots, Google's Gemini*). This ensures that participants anchor their responses to the current generation of AI assistants relevant to the study's context.

- **Final Item (used in this study):** "How often have you used any types of chatbots (e.g., ChatGPT, WhatsApp Chatbots, Google's Gemini) before?"

- **Measurement:** 7-point frequency scale (1 = Never, 7 = Daily or almost daily)

5. References

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